

Liam Tanner

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TECHNICAL SKILLS

Languages: Python, C++, C, SQL, JavaScript, TypeScript, Linux Bash
Backend & Systems: FastAPI, PostgreSQL, MongoDB, RESTful APIs, ETL Data Pipelines, ROS 2, CUDA
AI & Machine Learning: PyTorch, Large Language Models (LLMs), RAG, Scikit-learn, Computer Vision
Web & Tools: React, Docker, Git, Agile Methodologies, AWS (Cloud Practitioner Certified)

EDUCATION

University of Victoria Victoria, BC
Bachelor of Software Engineering Sep 2021 to Aug 2026
• 4.00/4.00 GPA - 93% Avg (2025/2026) | Coursework: Software Architecture, Database Systems, Data Structures

EXPERIENCE

Software Developer Jun 2025 to Dec 2025
Natural Resources Canada Victoria, BC

- Designed and implemented automated ETL pipelines alongside a team of software engineers, transforming massive PostgreSQL datasets into SQLite and flatfile releases to reduce processing time from hours to seconds.
- Engineered a release-versioning system using SQL and encryption to securely encode and track metadata.
- Authored Linux Bash scripts to automate the extraction and repair of a legacy system without source code, streamlining the developer experience by instantiating functional local environments instantly.

Software Engineer Jan 2025 to Aug 2026
AUVIC - Autonomous Underwater Vehicles Club Victoria, BC

- Engineered a ROS 2 dynamic configuration pipeline to extract PID and Kalman filter values from YAML files, eliminating recompilation downtime and enabling real-time tuning.
- Resolved critical thruster saturation limits and engineered an autonomous navigation state machine, advancing to the semi-finals of the international RoboSub 2026 competition.

Research Assistant - Induced Seismicity Project Sep 2023 to Apr 2024
Natural Resources Canada Sidney, BC

- Engineered increasingly complex CO2 injection simulations using Python meshes and the OpenGeoSys finite element solver, becoming the first researcher to implement this data-driven workflow.
- Advanced the project to a real-world case study by engineering a 3-layer finite element simulation of the SECURE INGA disposal well using precise depth profiles and statistical material properties.

PROJECTS

AI Incident Orchestrator | *Python, PostgreSQL, LLMs, RAG* May 2026 to Aug 2026

- Engineered an AI orchestration pipeline utilizing the Groq API and PostgreSQL to autonomously diagnose and recommend mitigations for cybersecurity threats such as privilege escalation and lateral movement.
- Integrated modern AI protocols and Retrieval-Augmented Generation (RAG) flows to evaluate unstructured data and automate real-time incident response recommendations.

EfficientAD Anomaly Detector | *Python, PyTorch, Computer Vision* Mar 2026

- Engineered a visual anomaly detection pipeline in PyTorch by training the teacher network and resolving student algorithm bottlenecks to allow for 70,000 iterations in under 2 hours.
- Evaluated model quality on a diverse industrial dataset, achieving an AUROC score of 0.88 while optimizing for sub-millisecond inference latency.

Convolutional Engine | *C++, CUDA, Hardware Optimization* Apr 2026

- Engineered a high-performance image processing library in C++ and CUDA, demonstrating strong proficiency in low-level memory management and statically typed execution.
- Implemented shared memory tiling to minimize global memory bottlenecks and optimize hardware-level memory throughput for parallel processes.

Virtual Triage System | *React, TypeScript, Python, FastAPI, MongoDB* Sep 2024 to Nov 2024

- Engineered a full-stack web application utilizing React and a Python/FastAPI back-end to securely process patient symptoms and manage virtual triage queues.
- Designed and developed RESTful APIs with MongoDB to allow remote nurses to review patient data, assign priority levels, and broadcast real-time wait times in local testing environments.